MATCH-Net: Dynamic Prediction in Survival Analysis using Convolutional Neural Networks

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Abstract

Accurate prediction of disease trajectories is critical for early identification and timely treatment of patients at risk. Conventional methods in survival analysis are often constrained by strong parametric assumptions and limited in their ability to learn from high-dimensional data, while existing neural network models are not readily-adapted to the longitudinal setting. This paper develops a novel convolutional approach that addresses these drawbacks. We present MATCH-Net: a Missingness-Aware Temporal Convolutional Hitting-time Network, designed to capture temporal dependencies and heterogeneous interactions in covariate trajectories and patterns of missingness. To the best of our knowledge, this is the first investigation of temporal convolutions in the context of dynamic prediction for personalized risk prognosis. Using real-world data from the Alzheimer's Disease Neuroimaging Initiative, we demonstrate state-of-the-art performance without making any assumptions regarding underlying longitudinal or time-to-event processes—attesting to the model's potential utility in clinical decision support.

1 Introduction

In Alzheimer's disease—the annual cost of which exceeds $800 billion globally [1]—the effectiveness of therapeutic treatments is often limited by the challenge of identifying patients at early enough stages of progression for treatments to be of use. As a result, accurate and personalized prognosis in earlier stages of cognitive decline is critical for effective intervention and subject selection in clinical trials. Conventional statistical methods in survival analysis often begin by choosing explicit functions to model the underlying stochastic process [2–7]. However, the constraints—such as linearity and proportionality [8, 9] in the popular Cox model—may not be valid or verifiable in practice.

In spite of active research, a conclusive understanding of Alzheimer's disease progression remains elusive, owing to heterogeneous biological pathways [10, 11], complex temporal patterns [12, 13], and diverse interactions [14, 15]. Hence Alzheimer's data is a prime venue for leveraging the potential advantages of deep learning models for survival. Neural networks offer versatile alternatives by virtue of their capacity as general-purpose function approximators, able to learn—without restrictive assumptions—the latent structure between an individual's prognostic factors and odds of survival.

Contributions. Our goal is to establish a novel convolutional model for survival prediction, using Alzheimer's disease as a case study for experimental validation. Primary contributions are threefold: First, we formulate a *generalized framework* for longitudinal survival prediction, laying the foundation for effective cross-model comparison. Second, our proposal is uniquely designed to capitalize on longitudinal data to issue *dynamically updated* survival predictions, accommodating potentially informative patterns of *data missingness*, and combining *human input* with model predictions. Third, we propose methods for deriving clinically meaningful insight into the model's inference process. Finally, we demonstrate state-of-the-art results in comparison with comprehensive benchmarks.

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2 Problem formulation

Let there be N patients in a study, indexed $i \in \{1, ..., N\}$. Time is treated as a discrete dimension of fixed resolution $\delta > 0$. Each longitudinal datum consists of the tuple $(t, \mathbf{x}_{i,t}, s_{i,t})$, where $\mathbf{x}_{i,t}$ is the vector of covariates recorded at time t, and $s_{i,t}$ is the binary survival indicator. Let random variable $T_{i,\text{surv}}$ denote the time-to-event, $T_{i,\text{cens}}$ the time of right-censoring, and $T_i = \min\{T_{i,\text{surv}}, T_{i,\text{cens}}\}$. Per convention, we assume that censoring is not correlated with the eventual outcome [16–18]. Let the complete longitudinal dataset be given by $\mathbf{X} = \{\langle(t, \mathbf{x}_{i,t}, s_{i,t})\rangle_{t=0}^{t_i}\}_{i=1}^{N}$. Then we can define

$$\mathbf{X}_{i,t,w} = \langle(t', \mathbf{x}_{i,t'}, s_{i,t'})\rangle_{t' \in T} \quad \text{where} \quad T = \{t' : t - w \leq t' \leq t\} \tag{1}$$

to be the set of observations for patient i extending from time t into a width-w window of the past, where parameter w depends on the model under consideration. Given longitudinal measurements in $\mathbf{X}_{i,t,w}$, our goal is to issue risk predictions corresponding to length-τ horizons into the future. Formally, given a *backward-looking* historical window $(t - w, t]$, we are interested in the failure function for *forward-looking* prediction intervals $(t, t + \tau]$; that is, we want to estimate the probability

$$F_i(t + \tau | t, w) = \mathbb{P}(T_{i,\text{surv}} \leq t + \tau | T_{i,\text{surv}} > t, \mathbf{X}_{i,t,w}) \tag{2}$$

of event occurrence within each prediction interval. Observe that parameterizing the width of the historical window results in a generalized framework—for instance, a Cox model only utilizes the most recent measurement; that is, $w = 1$. At the other extreme, recurrent models may consume the entire history; that is, $w = t$. As we shall see, the best performance is in fact obtained via a flexible intermediate approach—that is, by learning the optimal width of a sliding window of history.

3 MATCH-Net

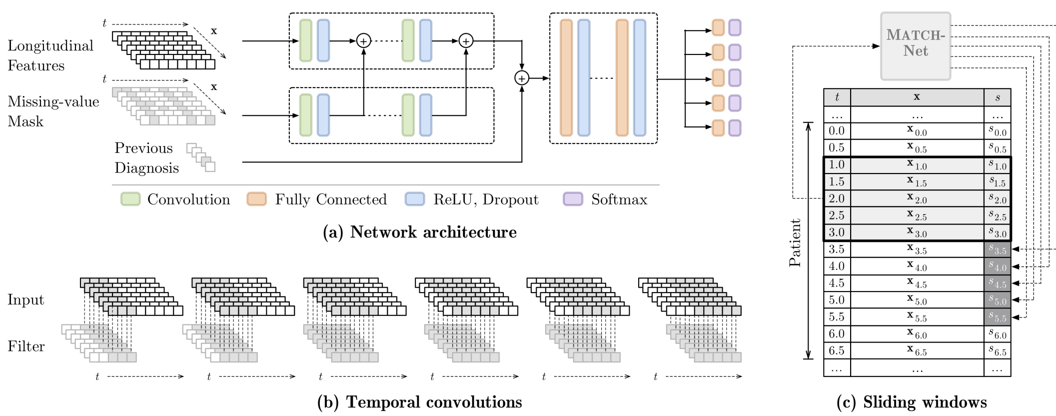

Figure 1: (a) The MATCH-Net architecture, with $\tau_{\text{max}} = 5\delta$. (b) Illustration of temporal convolutions acting over feature channels. (c) The longitudinal context within which MATCH-Net operates, as well as the network's prediction targets in association with the sliding window input mechanism.

We propose MATCH-Net: a Missingness-Aware Temporal Convolutional Hitting-time Network, innovating on current approaches in three respects. **Dynamic prediction**: Existing deep learning models issue prognoses on the basis of information from a single time point [19–29], potentially discarding valuable information in the presence of longitudinal data. We investigate *temporal convolutions* in capturing heterogeneous representations of temporal dependencies among observed covariates, enabling truly dynamic predictions on the basis of historical information. **Informative missingness**: Current survival methods rely on the common assumption that the timing and frequency of covariate measurements is uninformative [18, 30]. By contrast, our model is *missingness-aware*—that is, we explicitly account for informative missingness by learning correlations between patterns of missingness and disease progression. **Human input**: Instead of issuing predictions solely on the basis of quantitative clinical measurements, we optionally incorporate clinicians' most recent diagnoses of patient state into model estimates to examine the incremental informativeness of *subjective* input. Each innovation is a source of gain in performance (see Section 4; further details in the appendix).

MATCH-Net accepts as input a *sliding-window* of observed covariates $\mathbf{X}_{i,t,w}$, as well as a parallel binary mask of missing-value indicators $\mathbf{Z}_{i,t,w}$ taking on values of one to denote missing covariate measurements. In addition, the network optionally accepts a one-hot vector $\mathbf{r}_{i,t}$ describing the most recent clinician diagnosis of Alzheimer's disease progression. Starting from the base of the network, the convolutional block first learns representations of longitudinal covariate trajectories by extracting local features from temporal patterns. After each layer, filter activations from the auxiliary branch are concatenated with those in the main branch. Then, the fully-connected block captures more global relationships by combining local information. Finally, the output layers produce failure estimates

$$\hat{\mathbf{y}}_{i,t} = [\hat{F}_i(t+\delta|t,w), ..., \hat{F}_i(t+\tau_{\text{max}}|t,w)] \tag{3}$$

for pre-specified prediction intervals, where τ_{max} is the maximal horizon desired. This *convolutional dual-stream* architecture explicitly captures representations of temporal dependencies within each stream, as well as between covariate trajectories and missingness patterns in association with disease progression. This accounts for the potential informativeness of both *irregular* sampling (*i.e.* intervals between consecutive clinical visits may vary) and *asynchronous* sampling (*i.e.* not all features are measured at the same time) [31, 32]. This also encourages the network to distinguish between actual measurements and imputed values, reducing sensitivity to specific imputation methods chosen. The final architecture uses convolutions of length $l = 5\delta$, with more filters per layer in the main branch.

Loss function. With the preceding notation, the negative log-likelihood of a single empirical result $s_{i,t+\tau}$ and model estimate $\hat{F}_i(t+\tau|t,w)$ in relation to some input window $\mathbf{X}_{i,t,w}$ is given by

$$l_{i,t,\tau}(\boldsymbol{\theta}) = -[s_{i,t+\tau}\log\hat{F}_i(t+\tau|t,w) + (1-s_{i,t+\tau})\log(1-\hat{F}_i(t+\tau|t,w))] \tag{4}$$

where $\boldsymbol{\theta}$ denotes the parameters of the network. The total loss function is then computed to simultaneously take into account the quality of predictions for all prediction horizons τ, all times t available along each patient's longitudinal trajectory, and all patients $i \in \{1, ..., N\}$ in the survival dataset:

$$l(\boldsymbol{\theta}) = \frac{\delta^2}{\sum_{i=1}^{N}\sum_{j=1}^{t_i}\tau_i}\sum_{i=1}^{N}\sum_{j=1}^{t_i/\delta}\sum_{k=1}^{\tau_i/\delta}\alpha(i,(j-1)\delta,k\delta)\cdot l_{i,(j-1)\delta,k\delta} \tag{5}$$

where $\tau_i = \min\{t_i - t, \tau_{\text{max}}\}$ accounts for failure or right-censoring. This is a natural generalization of the log-likelihood in [24] to accommodate longitudinal survival. Weight function $\alpha(i,t,\tau)$ allows trading off the relative importance of different patients, time steps, and prediction horizons. First, this allows standardizing patient contributions with $\alpha(i,t,\tau) \propto 1/t_i$, thereby counteracting the bias against patients with shorter survival durations. Second, in the context of heavily imbalanced classes, this allows up-weighting positive instances—that is, input windows that correspond to actual failure.

4 Experiments

The Alzheimer's Disease Neuroimaging Initiative (ADNI) study data is a longitudinal survival dataset of per-visit measurements for 1,737 patients [1]. The data tracks disease progression through clinical measurements at ½-year intervals, including quantitative biomarkers, cognitive tests, demographics, and risk factors. Our objective is to predict the *first stable occurrence* of Alzheimer's disease for each patient. Further information on the dataset, preparation, and training can be found in the appendix.

Benchmarks. We evaluate MATCH-Net against both traditional longitudinal methods in survival analysis and recent deep learning approaches; the former includes Cox landmarking and joint modeling methods, and the latter includes static and dynamic multilayer perceptrons and recurrent neural network models. Performance is evaluated on the basis of the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC), both computed with respect to prediction horizons τ. Five-fold cross validation metrics are reported in Table 1.

Performance. MATCH-Net produces state-of-the-art results, consistently outperforming both conventional statistical and neural network benchmarks. Gains are especially apparent in AUPRC scores—improving on the MLP by an average of 15% and on joint models by 16% across all horizons, and by 27% and 26% for one-step-ahead predictions. To understand the sources of improvement, we observe a 4% gain in AUPRC from introducing the sliding window mechanism (MLP to S-MLP), a 9% gain from incorporating temporal convolutions (S-MLP to S-TCN), and a further 2% gain from accommodating informative missingness (S-TCN to MATCH-Net). In addition, *including* the most recent clinician diagnosis results in a further 17% gain (further details located in the appendix).

Table 1: Cross validation performance for $\tau_{\max} = 5\delta$ and $\delta = 1/2$ years: MATCH-Net *without* clinician input, sliding-window temporal convolutional networks (S-TCN) and multilayer perceptrons (S-MLP). Benchmarks include fully-convolutional networks (FCN) [33] adapted for sequence-based survival prediction, Disease Atlas (D-Atlas) [34], baseline recurrent neural networks (RNN) including GRUs and LSTMs, static multilayer perceptrons (MLP), as well as conventional statistical methods for survival analysis—including joint modeling (JM) and Cox landmarking (LM). Bold values indicate best performance, and asterisks next to benchmark results indicate statistically significant difference (p-value < 0.05) from MATCH-Net result. More detailed breakdown of gains are found the appendix.

	τ	MATCH-Net	S-TCN	S-MLP	FCN	D-Atlas	RNN	MLP	JM	LM
AUROC	0.5	**0.962**	0.961	0.959	0.954	0.959	0.949*	0.948*	0.913*	0.909*
	1.0	**0.942**	0.941	0.932	0.930	0.929	0.930	0.930	0.917*	0.914*
	1.5	**0.902**	**0.902**	0.897	0.895	0.892	0.891	0.890	0.881	0.878
	2.0	**0.909**	0.908	0.904	0.903	0.896	0.901	0.895	0.894	0.890
	2.5	**0.886**	0.884	0.881	0.883	0.884	0.883	0.874	0.883	0.878
AUPRC	0.5	**0.594**	0.580	0.500	0.536	0.517	0.464*	0.469*	0.473*	0.469*
	1.0	**0.513**	0.505	0.447	0.453	0.423	0.410*	0.435	0.415*	0.412*
	1.5	**0.373**	0.367	0.354	0.357	0.364	0.340	0.340	0.319	0.325
	2.0	**0.390**	0.380	0.364	0.375	0.352	0.355	0.359	0.362	0.367
	2.5	**0.384**	0.381	0.371	0.365	0.360	0.365	0.356	0.366	0.363

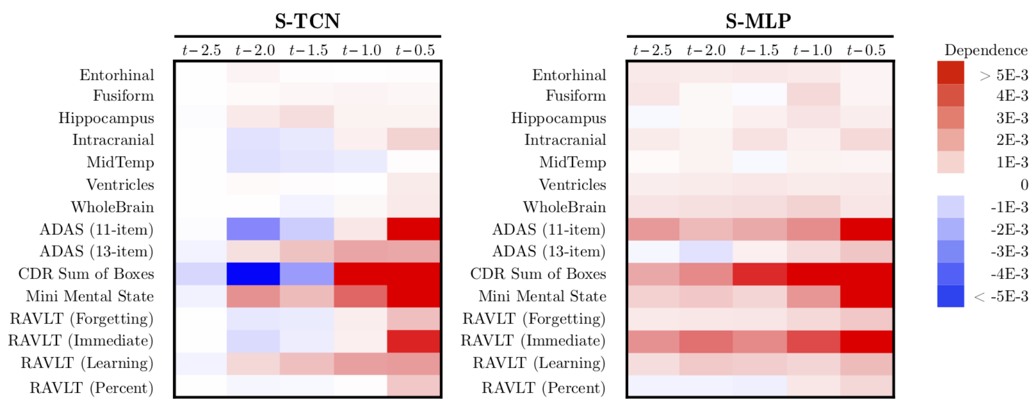

Figure 2: Average saliency map indicating feature and temporal influence, computed using slopes of partial dependence on sample of numerical features across sliding window ($w = 5\delta$; $\delta = 1/2$ years).

Visualization. From the preceding, we observe the largest gains by introducing convolutions. While this is consistent with our motivating hypothesis that convolutions are better able to capture temporal patterns, clinicians often desire a degree of transparency into the prediction process [35, 36]. We adopt the partial dependence approach in [37] to understand the input-output relationship, as well as examining the utility of convolutions. For each observed covariate d, we want to approximate how the estimated failure function varies based on the value of $\mathbf{x}^d_{t,w}$. We define the dependence

$$\mathbb{E}_{\mathbf{X}^{(-d)}_{t,w}}[\hat{F}(t+\tau|\mathbf{X}_{t,w})] \approx \frac{\delta}{\sum_{i=1}^{N} t_i} \sum_{i=1}^{N} \sum_{j=1}^{t_i/\delta} \hat{F}((j-1)\delta + \tau | \mathbf{x}^d_{(j-1)\delta,w}, \mathbf{X}^{(-d)}_{(j-1)\delta,w}) \tag{6}$$

where $\mathbf{x}^d_{t,w} \cup \mathbf{X}^{(-d)}_{t,w} = \mathbf{X}_{t,w}$. By evaluating Equation 6 on the values $\mathbf{x}^d_{t,w}$ present in the data, the influence of each covariate can be measured by estimating its slope. For a global picture of what impact each feature and time step has on the model's predictions, we compute the influence for all features to produce an average *saliency map* [38, 39] highlighting the effect of convolutional layers. All else equal, absent convolutions we see that having worse covariate values any time step almost invariably has an upward impact on risk (*i.e.* negative impact on survival). On the other hand, with temporal convolutions we see that having worse covariate values at earlier time steps may result in a downward impact on risk (*i.e.* positive impact on survival), suggesting that convolutions may better facilitate modeling relative movements (*e.g.* sudden declines) than simply paying attention to levels. Further visualizations for added perspective on input-output relationships are found in the appendix.

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